Product design process enhancement -

AI assisted design to code conversion



Context

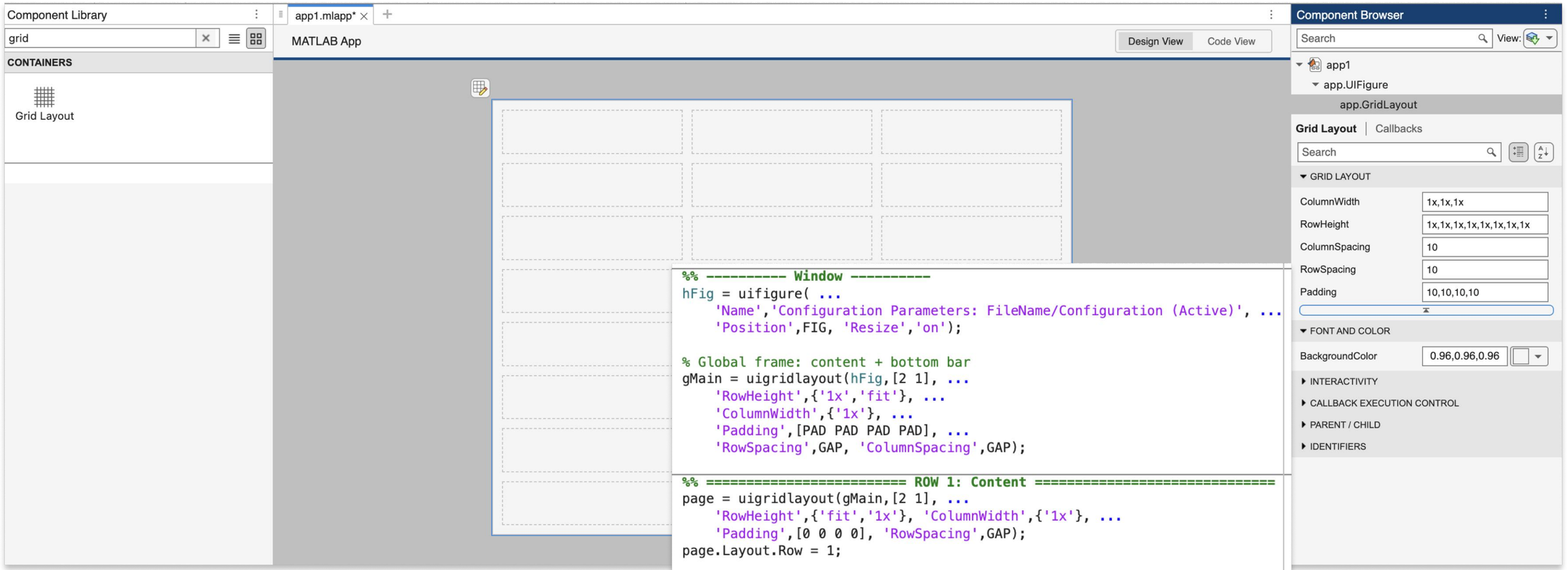
Before 2026 the design handover was

- confluence document,
- miro boards
- presentations
- Figma dev mode



Constraint

The MATLAB GUI is still built in the MATLAB language, based on the uigridlayout format.




```
%% ---------- Window ----------
hFig = uifigure( ...
    'Name','Configuration Parameters: FileName/Configuration (Active)', ...
    'Position',FIG, 'Resize','on');

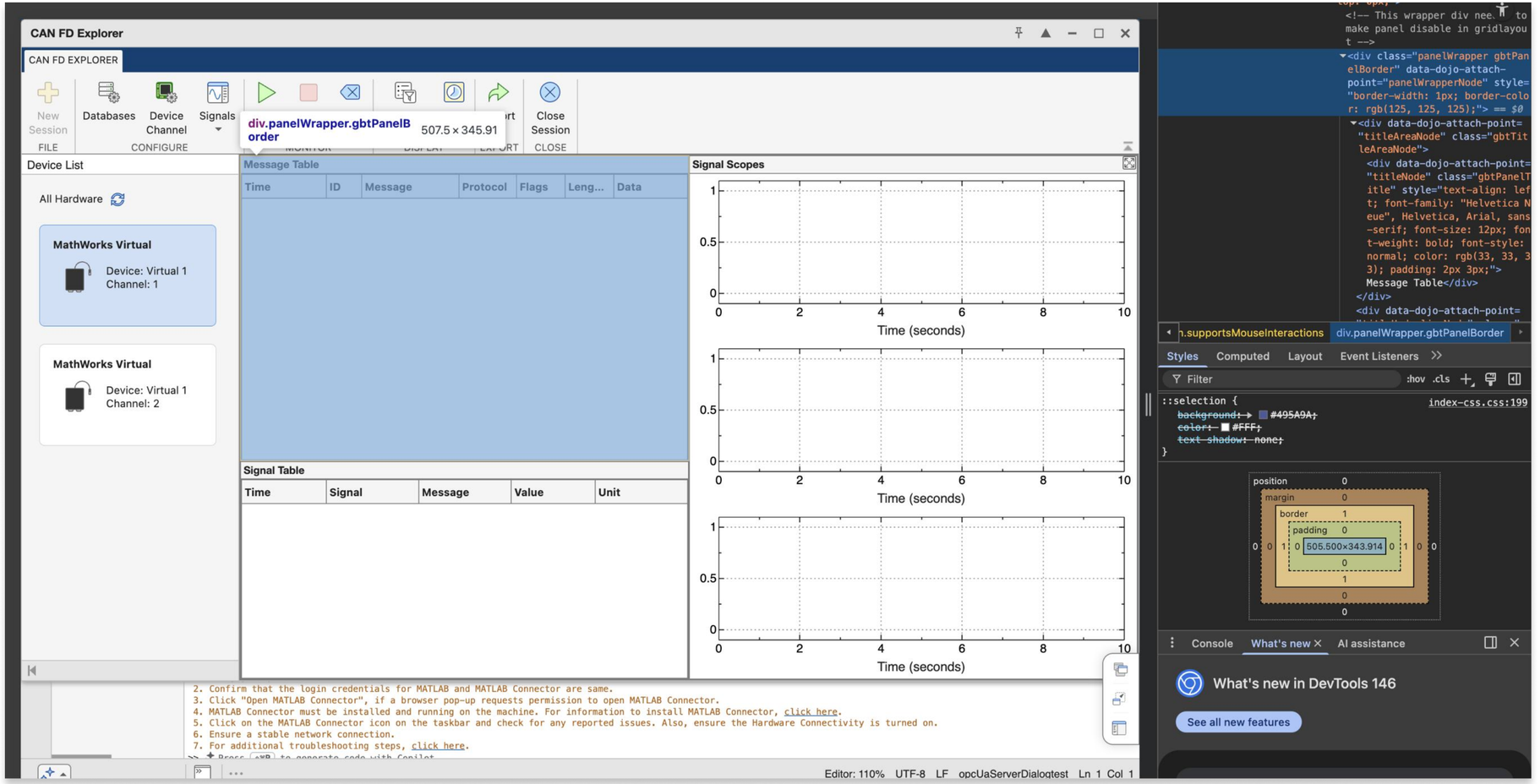
% Global frame: content + bottom bar
gMain = uigridlayout(hFig,[2 1], ...
    'RowHeight',{'1x','fit'}, ...
    'ColumnWidth',{'1x'}, ...
    'Padding',[PAD PAD PAD PAD], ...
    'RowSpacing',GAP, 'ColumnSpacing',GAP);
```

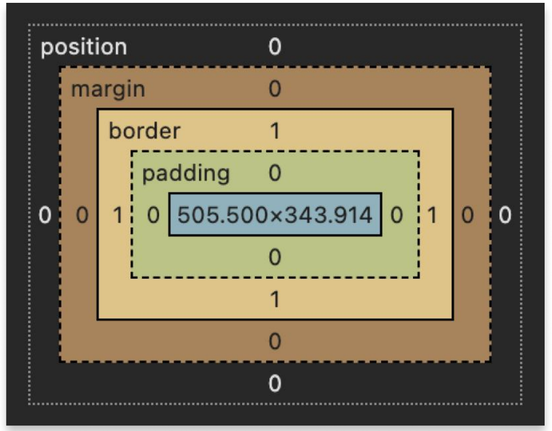
```
%% ======================== ROW 1: Content ================================
page = uigridlayout(gMain,[2 1], ...
    'RowHeight',{'fit','1x'}, 'ColumnWidth',{'1x'}, ...
    'Padding',[0 0 0 0], 'RowSpacing',GAP);
page.Layout.Row = 1;

% Search row
rowSearch = uigridlayout(page,[1 2], ...
    'RowHeight',{'fit'}, 'ColumnWidth',{'fit','1x'}, ...
    'Padding',[0 0 0 0], 'ColumnSpacing',GAP);
lblSearch = hdrLabel(rowSearch,'Search',false);
lblSearch.Layout.Row = 1; lblSearch.Layout.Column = 1;
efSearch = uieditfield(rowSearch,'text','Placeholder','Search','Value','');
efSearch.Layout.Row = 1; efSearch.Layout.Column = 2;
```

Constraint

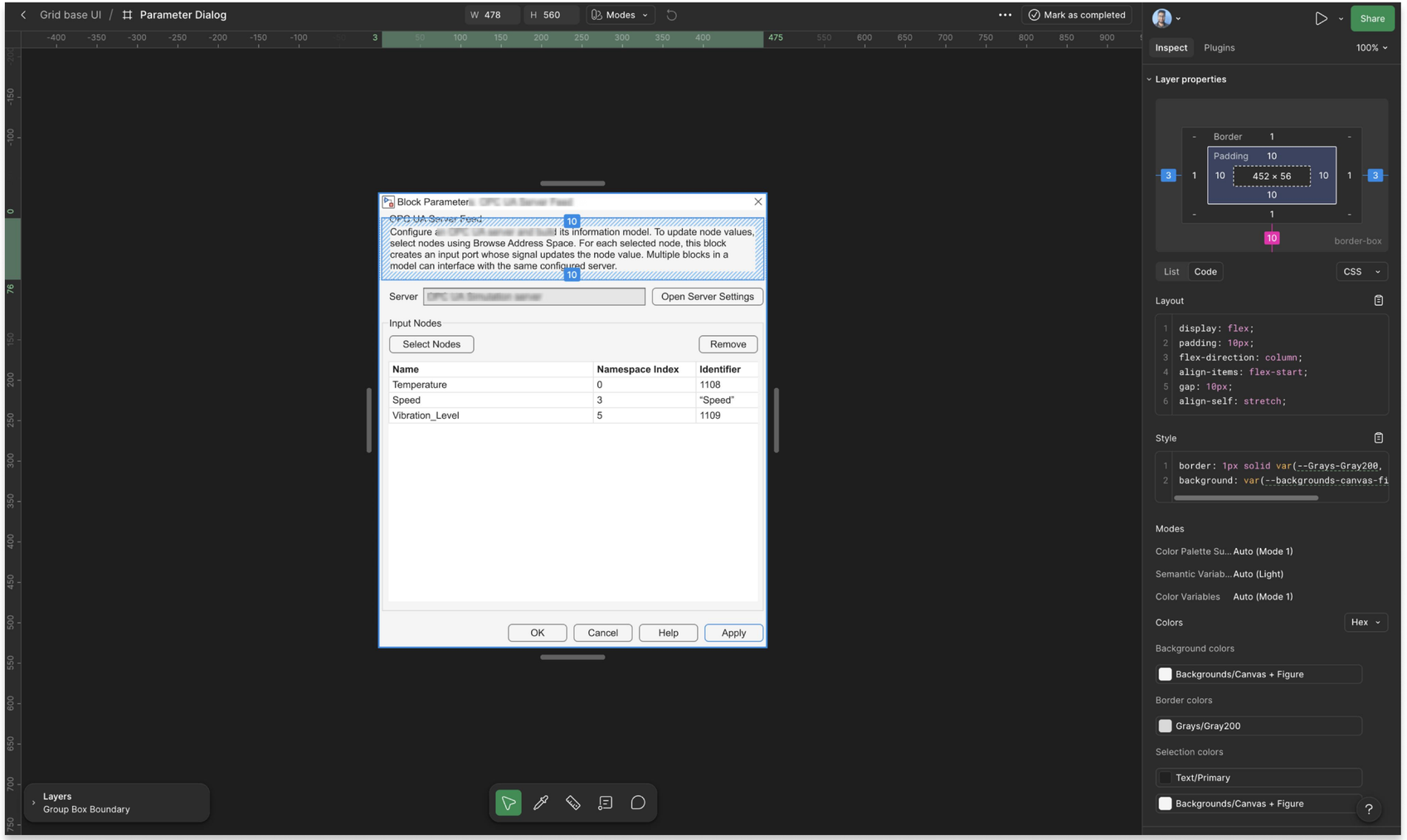
But web SaaS is HTML, CSS box model based





Constraint

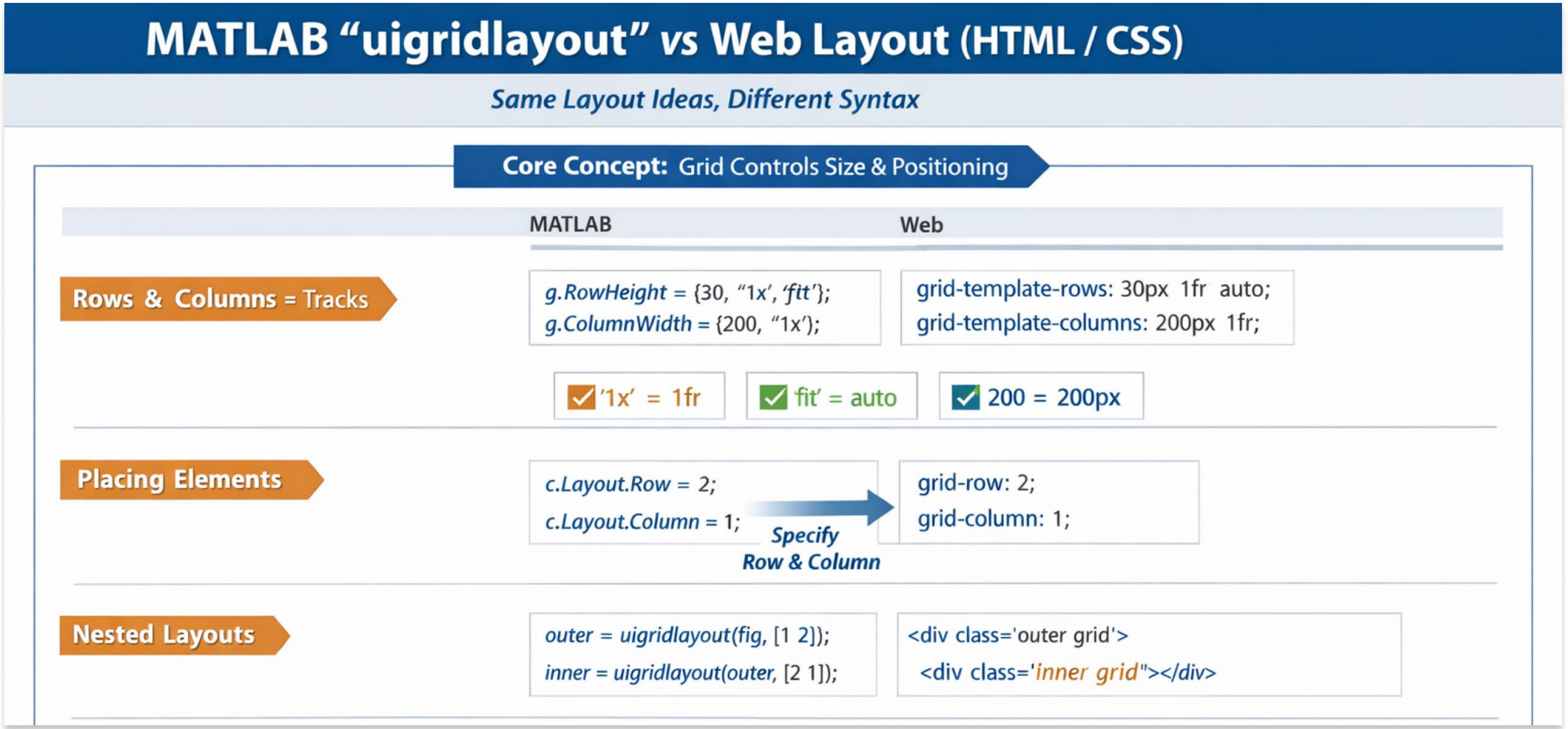
Also Figma dev mode mode also is HTML, CSS box model



Share
Inspect
Plugins
100%
Layer properties
Border 1
Padding 10
452 × 56
border-box
List
Code
CSS
Layout
display: flex;
padding: 10px;
flex-direction: column;
align-items: flex-start;
gap: 10px;
align-self: stretch;
Style
border: 1px solid var(--Grays-Gray200,
background: var(--backgrounds-canvas-fi
Modes
Color Palette Su... Auto (Mode 1)
Semantic Variab... Auto (Light)
Color Variables Auto (Mode 1)

Approach

Drawing parallels and creating standards around it



Solution

Figma UI to MATLAB code in minutes

Ai assisted UI grid that translates HTML, CSS box model to MATLAB uigridlayout

- used the internal coding guide document as a skill document
- leveraged MS-copilot to generate MATLAB code from Figma designs

Outcome

- 70% reduction in coding time for MATLAB uigrid based interfaces
- Enhanced UX-Dev collaboration

